

BSDS (2024 Semester V): Regression Techniques

Midsem Part B

NAME :

ROLL :

Time: 60 minutes

Maximum attainable marks: 25

Throughout this question paper, we will consider the multiple linear regression model with response variable y and a list of regressors/covariates $\{x_1, \dots, x_p\}$. Based on n observations the model is

$$\underline{Y} = X\underline{\beta} + \underline{\varepsilon}, \quad \text{where} \quad \underline{Y} = \begin{bmatrix} Y_1 \\ \vdots \\ Y_n \end{bmatrix}, \quad X = [\underline{1} \quad \underline{x}_{(1)} \quad \dots \quad \underline{x}_{(p)}], \quad \underline{\beta} = \begin{bmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_p \end{bmatrix}, \quad \text{and} \quad \underline{\varepsilon} = \begin{bmatrix} \varepsilon_1 \\ \vdots \\ \varepsilon_n \end{bmatrix}. \quad (1)$$

We assume that $\underline{\varepsilon} \sim N(0, \sigma^2 I_n)$.

Q.1 Consider the regression model in (1), and let the design matrix satisfy the following orthogonality property:

$$\underline{1}^\top \underline{x}_{(j)} = 0, \quad \text{and} \quad \underline{x}_{(k)}^\top \underline{x}_{(j)} = 0, \quad \text{for all } k \neq j, \quad j, k = 1, \dots, p.$$

Suppose we are interested in finding simultaneous confidence intervals for β_1 and β_2 , and we want to take a Bonferroni-type approach, where we inflate the confidence coefficients of individual confidence intervals to $1 - \alpha/r$ (from $1 - \alpha$), $\alpha > 0$, $r > 0$. Find the value of r and the confidence band $[L_j(\underline{Y}, \alpha), U_j(\underline{Y}, \alpha)]$, $j = 1, 2$, so that the confidence criterion is met exactly, i.e.,

$$P_{\underline{\beta}}(L_1(\underline{Y}, \alpha) < \beta_1 \leq U_1(\underline{Y}, \alpha), L_2(\underline{Y}, \alpha) < \beta_2 \leq U_2(\underline{Y}, \alpha)) = (1 - \alpha), \quad \forall \underline{\beta}.$$

[7]

To find a confidence interval for β_i , take $\underline{a}_i = \begin{pmatrix} 0 \\ \underline{e}_i \end{pmatrix}$, $i=1,2$.

Now the individual confidence interval with confidence coefficient $(1 - \alpha/n)$ is :

$$P_{\underline{\beta}} \left(\underline{a}_i^\top \underline{\beta} \in \underline{a}_i^\top \hat{\underline{\beta}}_{ols} \pm t_{\frac{\alpha}{2n}; n-p-1} \sqrt{\frac{RSS}{n-p-1}} \underline{a}_i^\top (X^\top X)^{-1} \underline{a}_i \right) = (1 - \alpha/n), \quad \forall \underline{\beta} \quad i=1,2.$$

$$\text{Here } \underline{a}_i^\top \underline{\beta} = \beta_i, \quad \underline{a}_i^\top \hat{\underline{\beta}}_{ols} = \hat{\beta}_{i,ols}, \quad \underline{a}_i^\top (X^\top X)^{-1} \underline{a}_i = \|\underline{x}_{(i)}\|^{-2}, \quad i=1,2$$

To obtain SCI, consider $A = \begin{pmatrix} \tilde{a}'_1 \\ \tilde{a}'_2 \end{pmatrix}$, then

$$A\hat{\beta}_{OLS} \sim N(A\tilde{\beta}, \sigma^2 A(X^T X)^{-1} A^T)$$

$$\text{Here } A\hat{\beta}_{OLS} = \begin{pmatrix} \hat{\beta}_{1,OLS} \\ \hat{\beta}_{2,OLS} \end{pmatrix}; \quad A\tilde{\beta} = \begin{pmatrix} \beta_1 \\ \beta_2 \end{pmatrix}; \quad A(X^T X)^{-1} A^T = \begin{bmatrix} 0 & I_2 & 0 \end{bmatrix} \begin{pmatrix} \frac{1}{n} & 0 & \dots & 0 \\ 0 & \|\tilde{x}_{(1)}\|^2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \|\tilde{x}_{(2)}\|^2 \end{pmatrix} \begin{bmatrix} 0^T \\ \tilde{I}_2 \\ 0 \end{bmatrix}$$

$$= \begin{pmatrix} \|\tilde{x}_{(1)}\|^2 & 0 \\ 0 & \|\tilde{x}_{(2)}\|^2 \end{pmatrix}$$

So, $\hat{\beta}_{1,OLS}$ and $\hat{\beta}_{2,OLS}$ are independently distributed.

$$\begin{aligned} \text{Thus } P_{\tilde{\beta}} \left(\beta_1 \in \hat{\beta}_{1,OLS} \pm t_{\frac{\alpha}{2n}; n-p-1} \sqrt{\frac{RSS}{n-p-1}} \|\tilde{x}_{(1)}\|^{-2}, \beta_2 \in \hat{\beta}_{2,OLS} \pm t_{\frac{\alpha}{2n}; n-p-1} \sqrt{\frac{RSS}{n-p-1}} \|\tilde{x}_{(2)}\|^{-2} \right) \\ = P_{\tilde{\beta}} \left(\beta_1 \in \hat{\beta}_{1,OLS} \pm t_{\frac{\alpha}{2n}; n-p-1} \sqrt{\frac{RSS}{n-p-1}} \|\tilde{x}_{(1)}\|^{-2} \right) \cdot P_{\tilde{\beta}} \left(\beta_2 \in \hat{\beta}_{2,OLS} \pm t_{\frac{\alpha}{2n}; n-p-1} \sqrt{\frac{RSS}{n-p-1}} \|\tilde{x}_{(2)}\|^{-2} \right) \\ = (1 - \frac{\alpha}{n})^2 \end{aligned}$$

We need to choose n such that

$$(1 - \frac{\alpha}{n})^2 = (1 - \alpha) \Leftrightarrow 1 - \frac{2\alpha}{n} + \frac{\alpha^2}{n^2} = 1 - \alpha$$

$$\Leftrightarrow \frac{\alpha^2}{n^2} = \alpha \left(\frac{2}{n} - 1 \right)$$

$$\Leftrightarrow \alpha = n^2 \left(\frac{2}{n} - 1 \right) \quad [\because \alpha \neq 0]$$

$$\Leftrightarrow n^2 - 2n + \alpha = 0$$

$$\text{So, } n = \frac{2 \pm \sqrt{4 - 4\alpha}}{2} = 1 \pm \sqrt{1 - \alpha}$$

Since $1 - \frac{\alpha}{n} > 0$, so $n > \alpha$. But $n = 1 - \sqrt{1 - \alpha} < \alpha$, so we should only consider the solution $n = 1 + \sqrt{1 - \alpha}$.

Q.2 Consider the normal linear regression setup as described in equation (1). Suppose we *know* a future value for the set of predictors, say $\underline{x}_0$, and we are interested in finding simultaneous prediction intervals for k -many future responses, say $\{Y_{0,1}, \dots, Y_{0,k}\}$, where $Y_{0,j} = \mu_0 + \varepsilon_{0,j}$ with $\mu_0 = \underline{x}_0^\top \beta$ and $j = 1, \dots, k$. Towards that, *under the assumptions* $\underline{\varepsilon}_0 = (\varepsilon_{0,1}, \dots, \varepsilon_{0,k})^\top \sim N_k(\underline{0}, \sigma^2 I)$ and $\underline{\varepsilon}_0 \perp \underline{\varepsilon}$, consider the following steps:

- First, find an *individual* confidence interval for the regression mean μ_0 , with confidence coefficient $1 - \alpha/4$.
- Next, find a *simultaneous* confidence interval for the k random errors $\{\varepsilon_{0,1}, \dots, \varepsilon_{0,k}\}$, with confidence coefficient at least $1 - 3\alpha/4$.
- Using the intervals obtained in (a)-(b), find a simultaneous prediction interval for $\{Y_{0,1}, \dots, Y_{0,k}\}$ with confidence coefficient at least $(1 - \alpha)$.

In each case, write the steps clearly.

[4 + 5 + 5]

(a) Since $\underline{x}_0$ is given, we find the individual confidence interval from the following:
 $\underline{x}_0^\top \hat{\beta}_{OLS} \sim N(\underline{x}_0^\top \beta, \sigma^2 \underline{x}_0^\top (X^\top X)^{-1} \underline{x}_0)$
 and $\frac{RSS}{\sigma^2} \sim \chi^2_{n-p-1}$ independently, as $\hat{\beta}_{OLS}$ and RSS are independent.

$$\text{So, } \frac{\underline{x}_0^\top \hat{\beta}_{OLS} - \underline{x}_0^\top \beta}{\sqrt{\frac{RSS}{n-p-1} \underline{x}_0^\top (X^\top X)^{-1} \underline{x}_0}} \sim t_{n-p-1} \quad \text{where } \underline{x}_0^\top \beta = \mu_0$$

$$\text{Thus, } P_{\beta} \left(\mu_0 \in \underline{x}_0^\top \hat{\beta}_{OLS} \pm t_{\frac{\alpha}{8}; n-p-1} \sqrt{\frac{RSS}{n-p-1} \underline{x}_0^\top (X^\top X)^{-1} \underline{x}_0} \right) = P_{\beta}(E_0) \quad (\text{let}) \\ = 1 - \frac{\alpha}{4}$$

(b) We know that,

$$\frac{\underline{\varepsilon}_0^\top \underline{\varepsilon}_0}{\sigma^2} \sim \chi^2_k$$

$$\text{By Scheffe's method, } \sup_{\underline{a} \neq \underline{0}} \frac{(\underline{a}^\top \underline{\varepsilon}_0)^2}{\|\underline{a}\|^2} = \underline{\varepsilon}_0^\top \underline{\varepsilon}_0$$

$$\text{Also, as } RSS \perp \underline{\varepsilon}_0, \text{ we have } \sup_{\underline{a} \neq \underline{0}} \frac{(\underline{a}^\top \underline{\varepsilon}_0)^2}{\|\underline{a}\|^2} \frac{n-p-1}{k} \sim F_{k, n-p-1}$$

$$\text{Thus, } P \left[|\underline{a}^\top \underline{\varepsilon}_0| \leq \|\underline{a}\| \sqrt{\frac{k}{n-p-1} F_{\frac{3\alpha}{4}; k, n-p-1}} \quad \forall \underline{a} \neq \underline{0} \right] = \left(1 - \frac{3\alpha}{4}\right)$$

In particular, taking $\underline{a} = \underline{e}_j$, $j=1,2,\dots,k$ so that $\|\underline{a}\|=1$,

$$P\left(|\varepsilon_{0,j}| \leq \sqrt{\frac{k \text{RSS}}{n-p-1}} F_{\frac{3\alpha}{4}; k, n-p-1} \quad \forall j=1,2,\dots,k\right) = P(E_i) \text{ (let)} \\ \geq 1 - \frac{3\alpha}{4}$$

(c) Combining (a) and (b) we get,

$$P[E_0 \cap E_i] = P(E_0) + P(E_i) - P(E_0 \cup E_i) \geq \left(1 - \frac{\alpha}{4}\right) + \left(1 - \frac{3\alpha}{4}\right) - 1 \\ = 1 - \alpha.$$

So the simultaneous prediction interval is :

$$P\left[\mu_0 + \varepsilon_{0,j} \in \underline{x}_0^T \hat{\underline{\beta}}_{OLS} \pm \hat{\sigma}_{UE} \left\{ t_{\frac{\alpha}{8}; n-p-1} \sqrt{\underline{x}_0^T (\underline{X}^T \underline{X})^{-1} \underline{x}_0} + \sqrt{k F_{\frac{3\alpha}{4}; k, n-p-1}} \right\}, \right. \\ \left. \forall j=1,2,\dots,k \right] \geq (1-\alpha).$$

where $\hat{\sigma}_{UE} = \sqrt{\frac{\text{RSS}}{n-p-1}}$.

Q.3 Consider the normal linear regression setup as described in equation (1). Suppose we want to test that, the regression coefficients associated with the covariates $\{x_1, \dots, x_q\}$ are the same, and that of $\{x_{q+1}, \dots, x_p\}$ are the same, $q < p$.

(a) Express the hypothesis (say, H_0) in $A\beta = c$ form.

(b) Find the residual sum of squares under H_0 , i.e., RSS_{H_0} .

(c) Hence, find the F -statistic and the exact distribution of $RSS_{H_0} - RSS$.

[3 + 4 + 4]

(a) $H_0: \beta_1 = \dots = \beta_q = \gamma_1$ and $\beta_{q+1} = \dots = \beta_p = \gamma_2$ where γ_1 and γ_2 are unknown.

$$\therefore H_0: A\beta = c \text{ where } A = \begin{bmatrix} 0 & I_p \end{bmatrix} \text{ and } c = \begin{pmatrix} \gamma_1 \cdot 1_q \\ \gamma_2 \cdot 1_{p-q} \end{pmatrix}$$

(b) Under H_0 , the linear model reduces to :

$$\begin{aligned} \underline{y} &= \beta_0 \underline{1} + \gamma_1 \underbrace{(\underline{x}_{(1)} + \dots + \underline{x}_{(q)})}_{\underline{z}_{(1)}} + \gamma_2 \underbrace{(\underline{x}_{(q+1)} + \dots + \underline{x}_{(p)})}_{\underline{z}_{(2)}} + \underline{\varepsilon} \\ &= \underline{Z}\underline{\gamma} + \underline{\varepsilon} \text{ where } \underline{Z} = \begin{bmatrix} \underline{1} & \underline{z}_{(1)} & \underline{z}_{(2)} \end{bmatrix} \text{ and } \underline{\gamma} = \begin{pmatrix} \beta_0 \\ \gamma_1 \\ \gamma_2 \end{pmatrix} \end{aligned}$$

Thus, under H_0 the RSS is

$$RSS_{H_0} = \underline{y}^T (I - P(\underline{Z})) \underline{y} = \underline{\varepsilon}^T (I - P(\underline{Z})) \underline{\varepsilon}.$$

(c) Without any constraint, the RSS is : $RSS = \underline{y}^T (I - P(\underline{x})) \underline{y}$

Hence $F = \frac{(RSS_{H_0} - RSS) / df_1}{RSS / n - p - 1}$ where df_1 is the degrees of freedom of the distⁿ of $RSS_{H_0} - RSS$.

$$\begin{aligned} \text{Under } H_0, \quad RSS_{H_0} - RSS &= (\underline{Z}\underline{\gamma} + \underline{\varepsilon})^T (P(\underline{x}) - P(\underline{Z})) (\underline{Z}\underline{\gamma} + \underline{\varepsilon}) \\ &= \underline{\varepsilon}^T (P(\underline{x}) - P(\underline{Z})) \underline{\varepsilon}, \text{ as } \mathcal{C}(\underline{Z}) \subseteq \mathcal{C}(\underline{x}) \\ &\Rightarrow P(\underline{x})\underline{Z} = P(\underline{Z})\underline{Z} = \underline{Z}. \end{aligned}$$

Also, $P(X) - P(Z)$ is idempotent and $\text{rank}[P(X) - P(Z)] = \text{tr}(P(X)) - \text{tr}(P(Z))$
 $= \text{rank}(X) - \text{rank}(Z)$

$$\text{Thus, } \frac{RSS_{H_0} - RSS}{\sigma^2} \sim \chi^2_{p-2} \quad \begin{array}{l} = p+1-3 \\ = p-2 \end{array}$$

$$\therefore df_1 = p-2.$$

$$\therefore \text{The F-statistic is, } F = \frac{(RSS_{H_0} - RSS) / (p-2)}{RSS / (n-p-1)}$$

$$= \left(\frac{n-p-1}{p-2} \right) \cdot \frac{\tilde{Y}^T (P(X) - P(Z)) \tilde{Y}}{\tilde{Y}^T (I - P(X)) \tilde{Y}}.$$

Rough work

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